

Lab: Build a System Model

Estimated duration: 40 minutes

Objectives

In this lab, you will:

- Construct a DFD to represent processes, data stores, and external interactions within a library system
- Design an ERD that defines entities, attributes, and relationships for an effective database schema
- Translate system functionality into structured models to support application development and system documentation

Before you begin

You should review the following content from Module 1 before starting this lab :

- **Video: Introduction to Data Flow Diagrams (DFDs)**
- **Video: Constructing Data Flow Diagrams (DFDs)**
- **Reading: Introduction to Entity-Relationship Diagrams (ERDs)**
- **Video: Developing ERDs for System Design**

Introduction

In this lab, you will build a system model for a library management system using two widely used techniques: the data flow diagram (DFD) and the entity-relationship diagram (ERD). These visual tools help represent how data moves through a system and how entities are related within a database. Through this hands-on lab, you will practice converting real-world requirements into structured models supporting system analysis and design.

Tools

You may use any diagramming tool to complete the tasks in this lab. Options include [Lucidchart](#) (the free version is recommended), [draw.io](#), [Microsoft Visio](#), or even pen and paper.

If you choose to upgrade Lucidchart to a paid plan, you will be responsible for any charges. New users can follow the reading [Getting Started with Lucidchart](#).

Deliverables:

- Create your context diagram, level 1 DFD, and ERD images

Scenario

You are tasked with designing a small library management system model.

The system must manage:

- **Borrowers** who can borrow and return books
- **Books** that are available for borrowing, with key details such as title, author, and ISBN
- **Librarians** who manage book borrowing and returns, and track overdue fines
- **Transactions** that record borrowing activities and ensure book tracking

Part 1: Create a DFD

Task 1: Context diagram

Why this diagram matters

A data flow diagram visualizes how data moves through a system. It clarifies processes, storage, and interactions with external entities.

Task:

Create a context diagram (Level 0 DFD) that shows the entire system as a single process with interactions to external entities.

Requirements:

- **External entities:** Borrower, Librarian
- **Main process:** Library Management System
- **Data flows:** Borrow Request, Return Request, Book Details, Fine Details, Borrower Details

Deliverable:

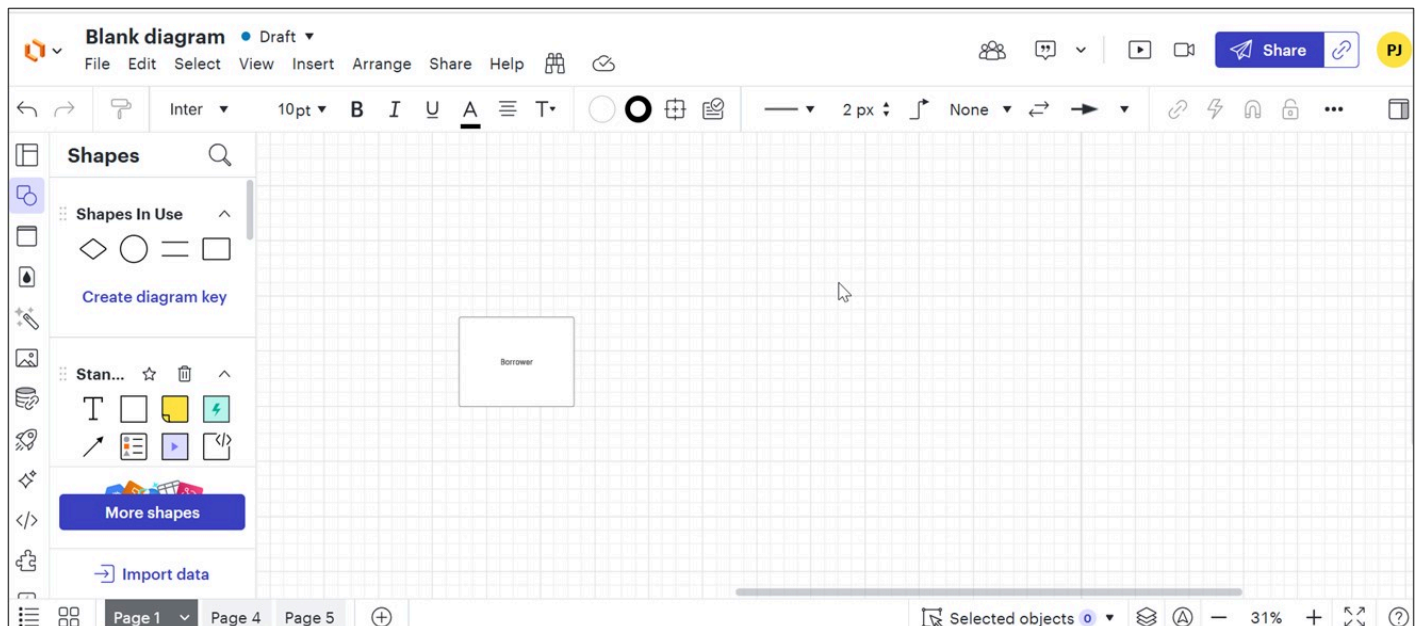
A context diagram with:

- One central process: "Library Management System"
- Two external entities: "Borrower" and "Librarian"
- Labeled data flows (for example, "Borrow Request" and "Book Details")

Steps:

Step 1: Create the "Borrower" box

1. Open your Lucidchart workspace to a blank diagram
2. In the Shapes panel on the left, locate the rectangle under Standard shapes
3. Click and drag the rectangle shape onto the canvas
4. Double-click inside the rectangle to edit the text, and type "Borrower"
5. Move the shape to the center of the canvas by clicking and dragging it

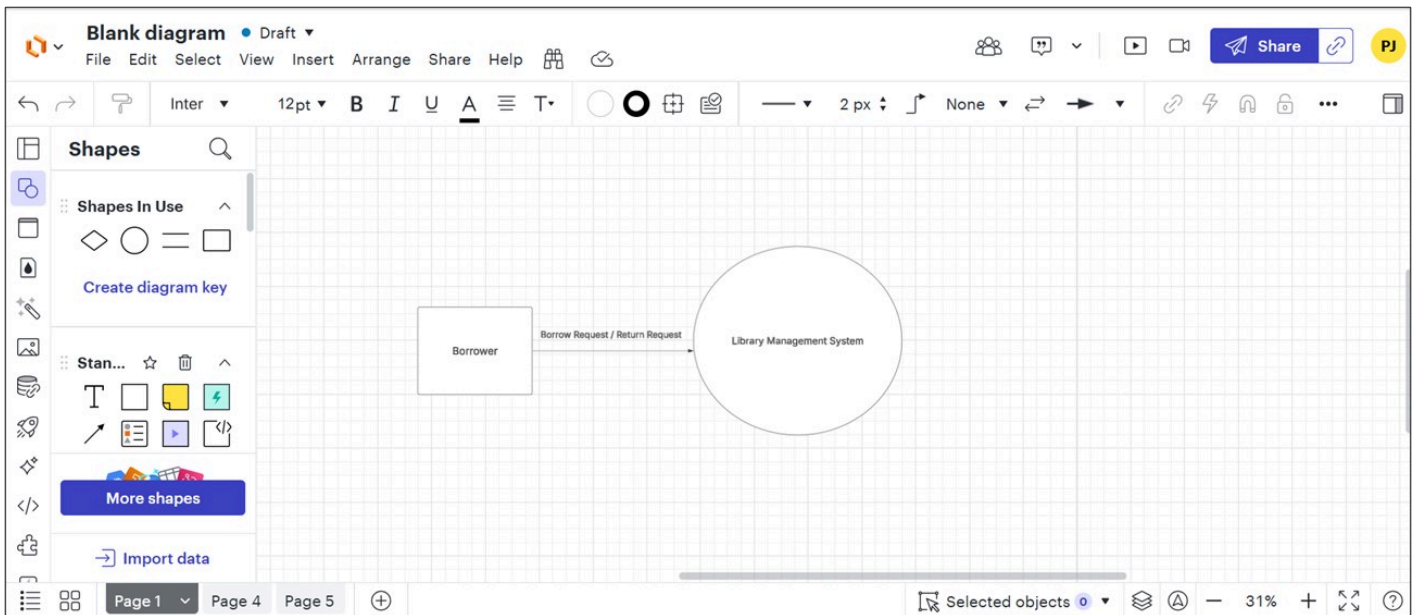


Step 2: Add the "Library Management System" process

1. Drag a circle from the Shapes panel onto the canvas
2. Double-click the circle and type "Library Management System"
3. Place it to the right of the "Borrower" rectangle

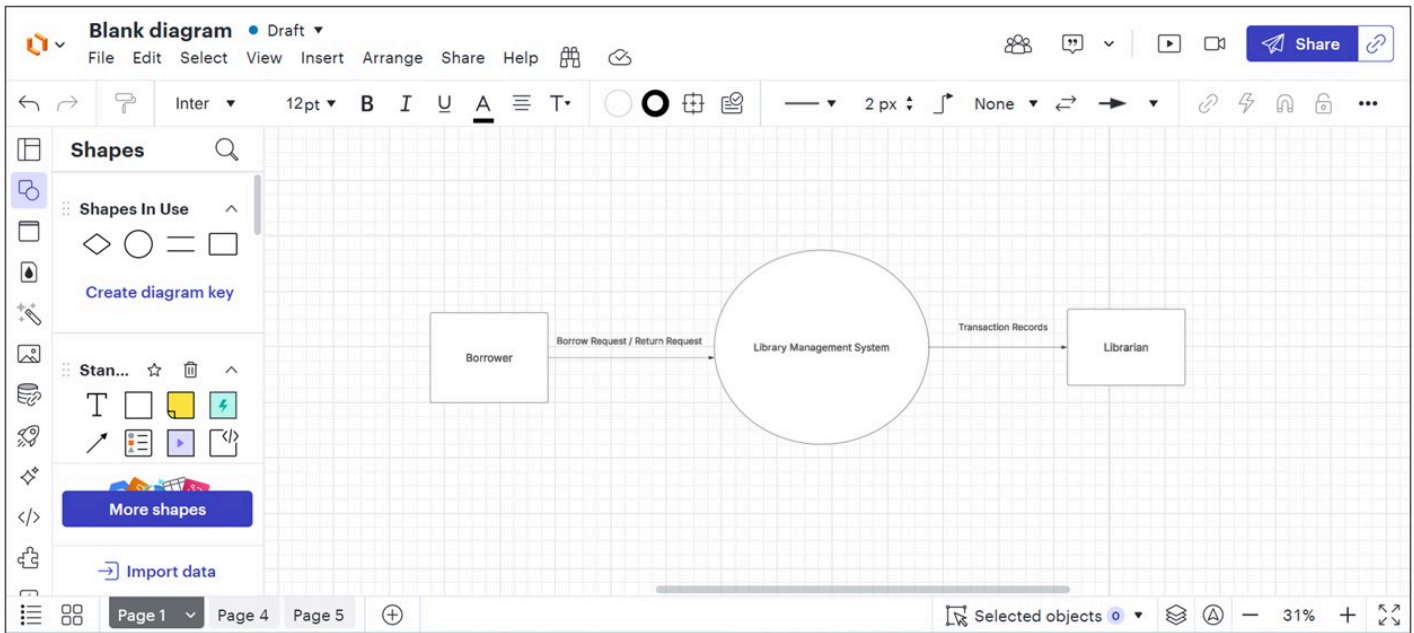
Step 3: Connect the entities

1. Hover over the "Borrower" box until you see connection points (blue dots)
2. Click and drag from one of the blue points toward the "Library Management System" circle to create a directional arrow
3. After the arrow appears, select it to add a text label
4. Type "Borrow Request / Return Request" to describe the data flow



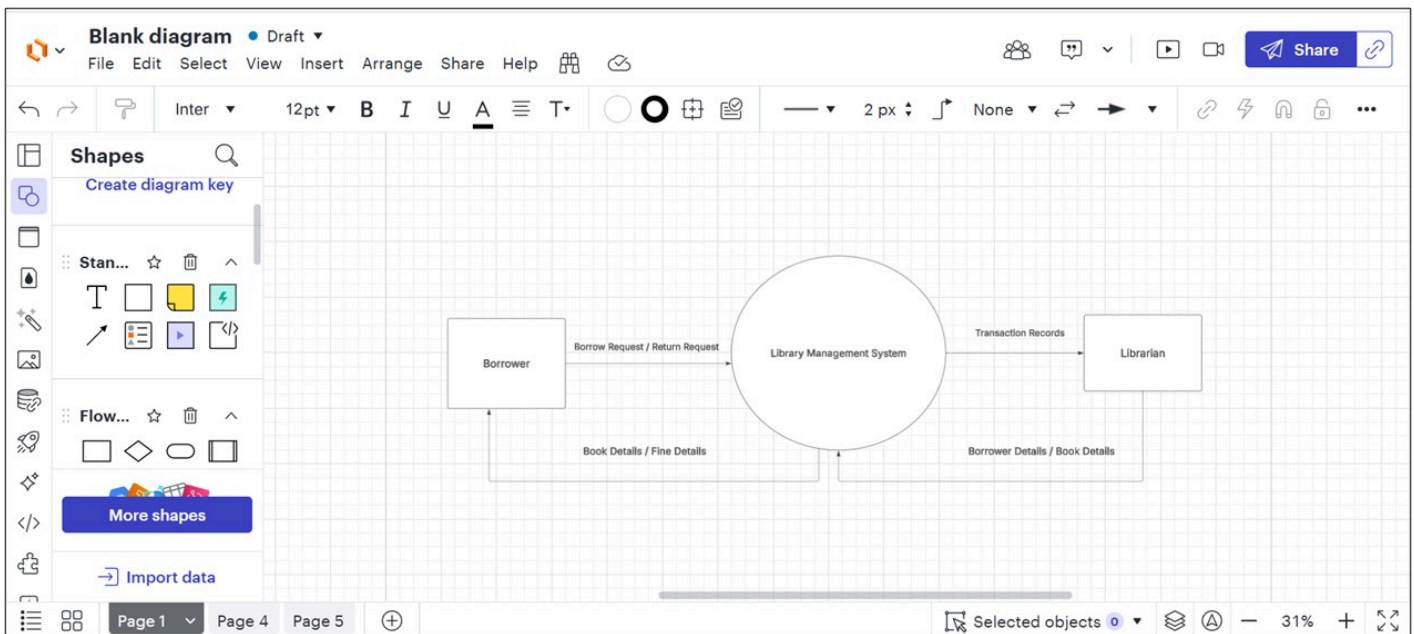
Step 4: Add the "Librarian" entity and connect it to the "Library Management System"

1. Drag another rectangle onto the canvas and place it to the right of the "Library Management System" circle
2. Label it "Librarian"
3. Hover over the "Library Management System" circle
4. Click and drag an arrow to the "Librarian" box
5. Label this arrow, "Transaction Records"



Step 5: Add connections from "Librarian" to "Library Management System" and from "Library Management System" to "Borrower"

1. Hover over the "Librarian" rectangle until connection points (blue dots) appear
2. Click and drag an arrow from the "Librarian" box to the "Library Management System" circle
3. Click the arrow line to add a label
4. Type **"Borrower Details / Book Details"** as the label
5. Similarly, add a connection arrow from the "Library Management System" circle to the "Borrower" rectangle
6. Type **"Book Details / Fine Details"** as the label



What this prepares you for:

Using a context diagram, you have now visualized the system at a high level. Next, you'll break it into detailed internal processes and data stores.

Task 2: Level 1 DFD

Use Lucidchart to decompose the context diagram into a Level 1 DFD to show internal processes, data stores, and data flows

Requirements:

- **Processes:** Register Borrower, Borrow Book, Return Book, Calculate Fine
- **Data stores:** Borrower Records, Book Inventory, Transaction Records
- **Data flows:** Show connections between processes, data stores, and external entities
- **Numbered processes:** Ensure that all processes are numbered (for example, 1.0, 2.0, and so on)

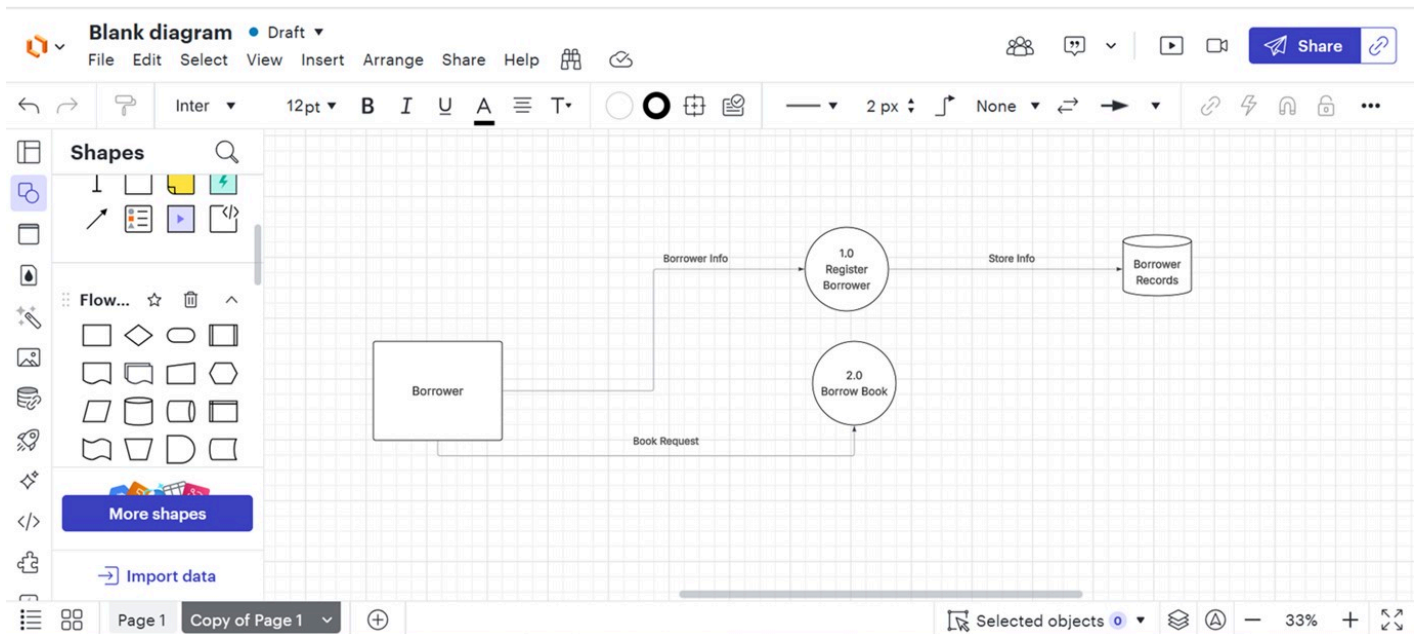
Deliverable:

A Level 1 DFD with:

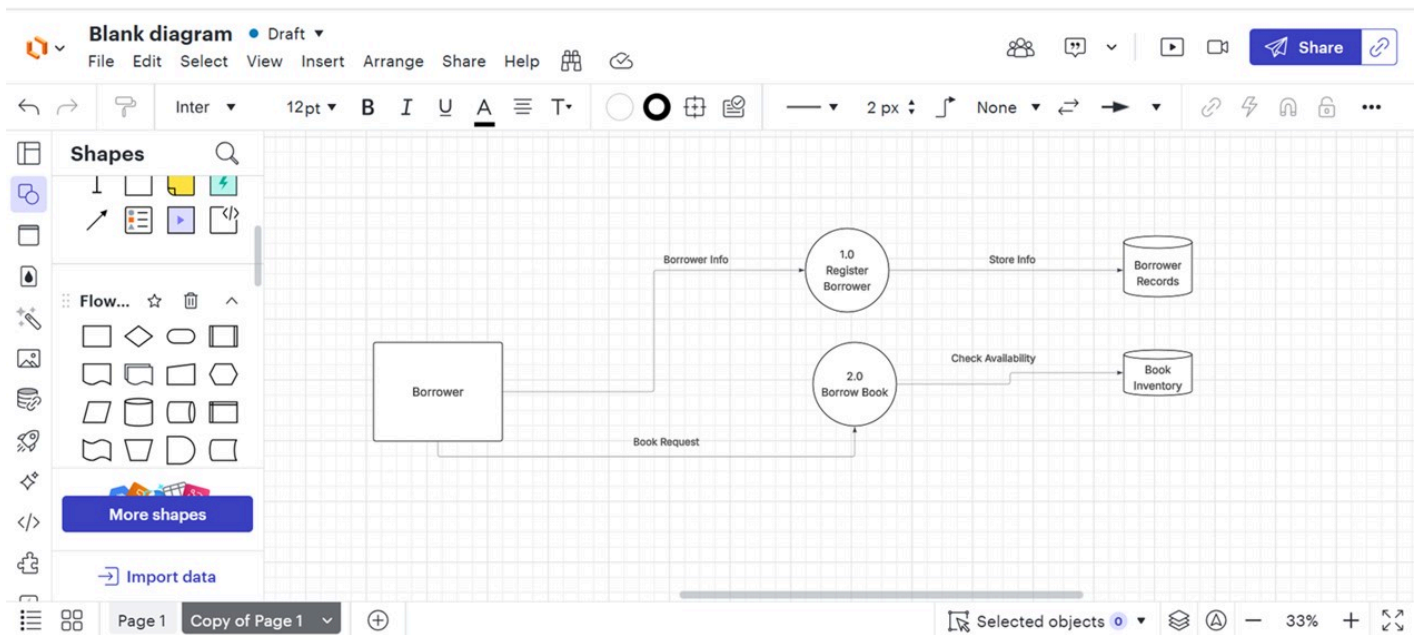
- Four processes: Register Borrower, Borrow Book, Return Book, Calculate Fine
- Three data stores: Borrower Records, Book Inventory, Transaction Records
- Labeled data flows connecting all processes, data stores, and external entities (Borrower, Librarian)

Steps:

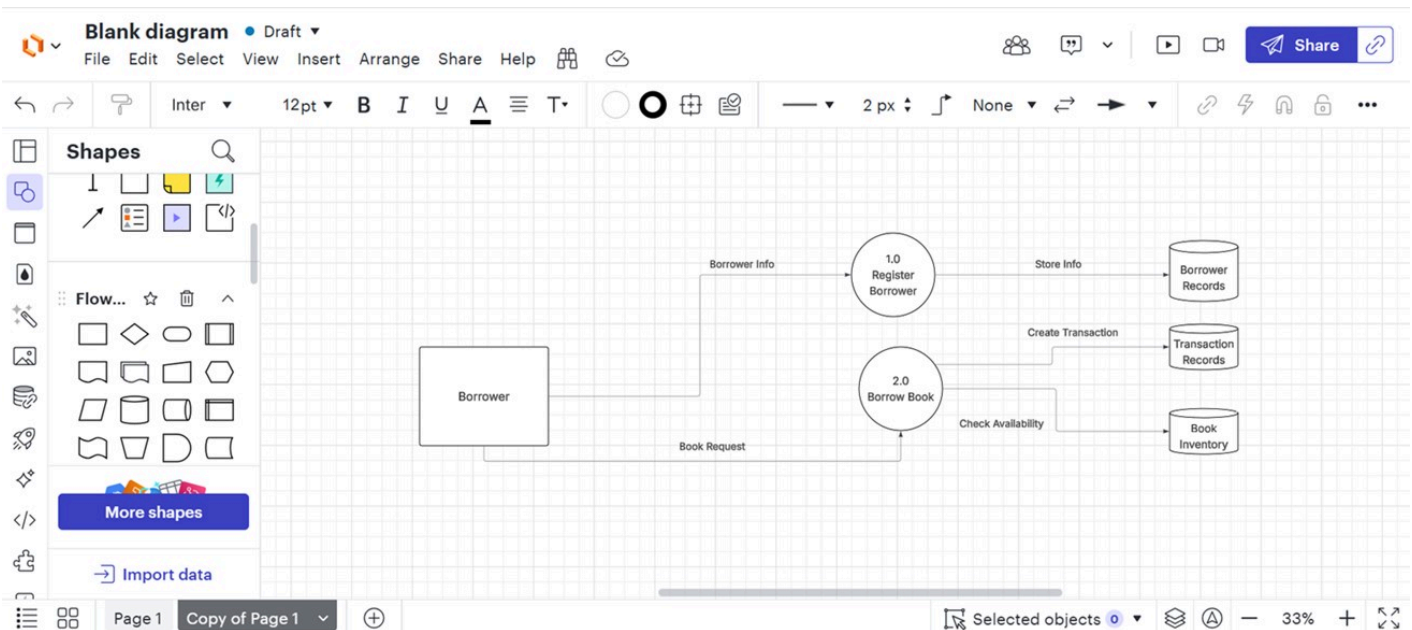
1. Break down the library management system into key processes



2. Identify data stores for Borrower Records, Book Inventory, and Transaction Records

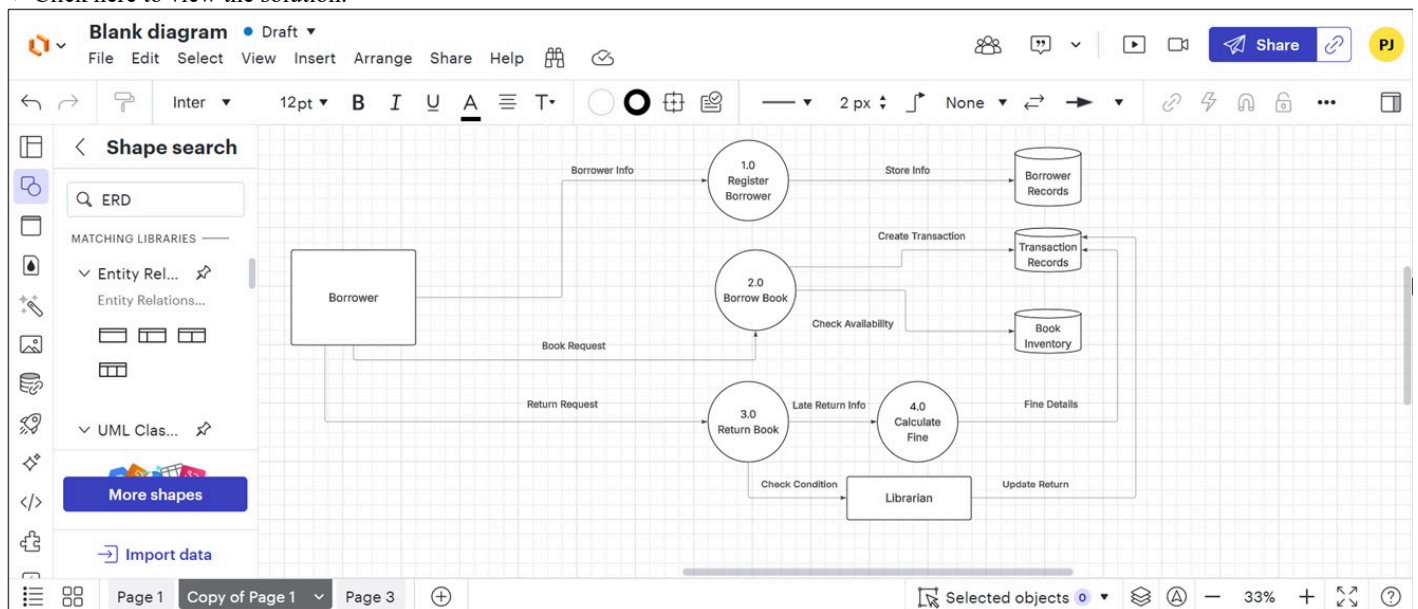


3. Map data flows between processes, stores, and external entities



4. Ensure that data flows are labeled clearly (for example, Borrower Info, Book Availability)

▼ Click here to view the solution:



What this prepares you for:

Using a context diagram, you have now visualized the system at a high level. Next, you'll break it into detailed internal processes and data stores.

Part 2: Create an ERD

Task 3: Design the ERD

Why this diagram matters

An ERD models the entities and relationships in a database. It ensures that your design captures all necessary elements for managing data effectively.

Task:

On Lucidchart, construct an ERD that defines the database schema for the library management system

Requirements:

- **Entities:**
 - Borrower, Book, Transaction, Librarian

- **Attributes:**

- **Borrower:** BorrowerID (primary key), Name, Email, Phone
- **Book:** ISBN (primary key), Title, Author, Status (Available/Borrowed)
- **Transaction:** TransactionID (primary key), BorrowerID (foreign key), ISBN (foreign key), BorrowDate, ReturnDate, FineAmount
- **Librarian:** LibrarianID (primary key), Name, Email

- **Relationships**

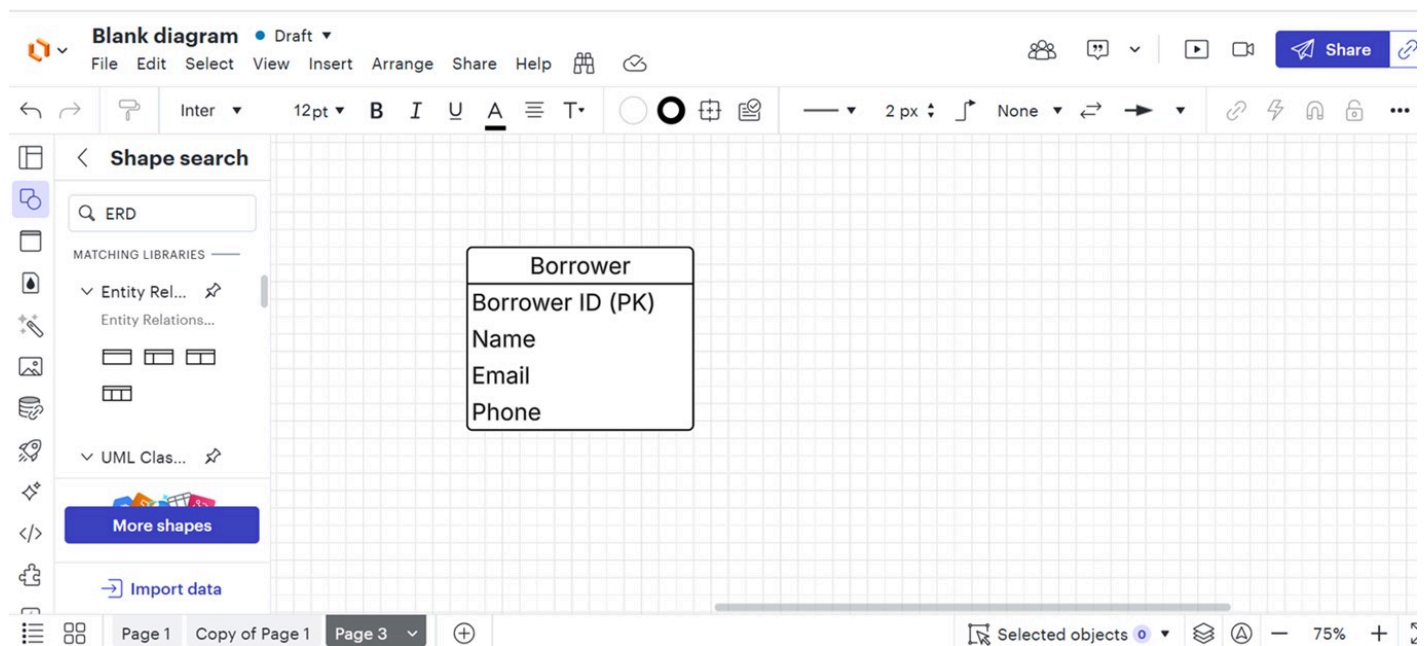
- Borrower borrows Books (through a Transaction)
- Librarian manages Transactions

- **Cardinality:**

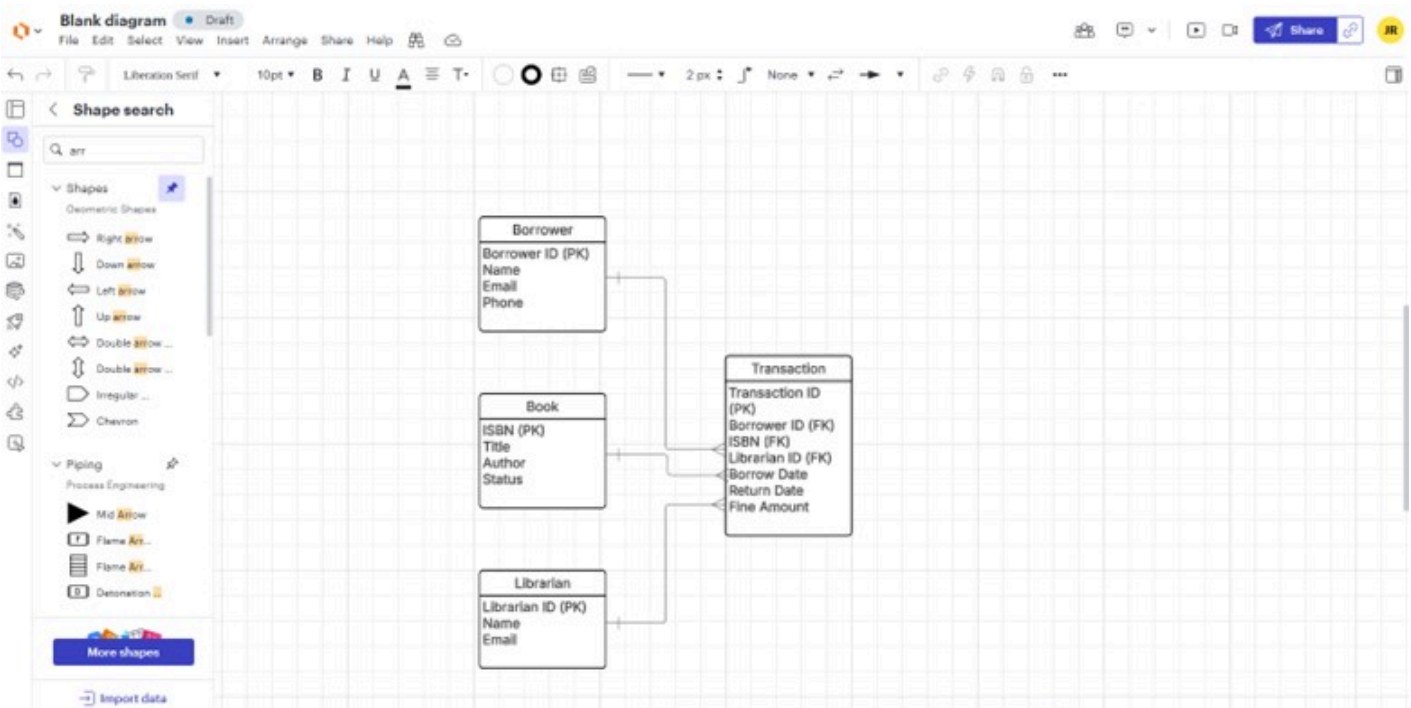
- A Borrower can have many transactions (1:N)
- A Book can appear in many transactions (1:N)
- A Librarian can manage many transactions (1:N)

Steps:

1. Identify entities and attributes
2. Assign primary and foreign keys (PK and FK)



3. Establish relationships between entities with appropriate cardinality



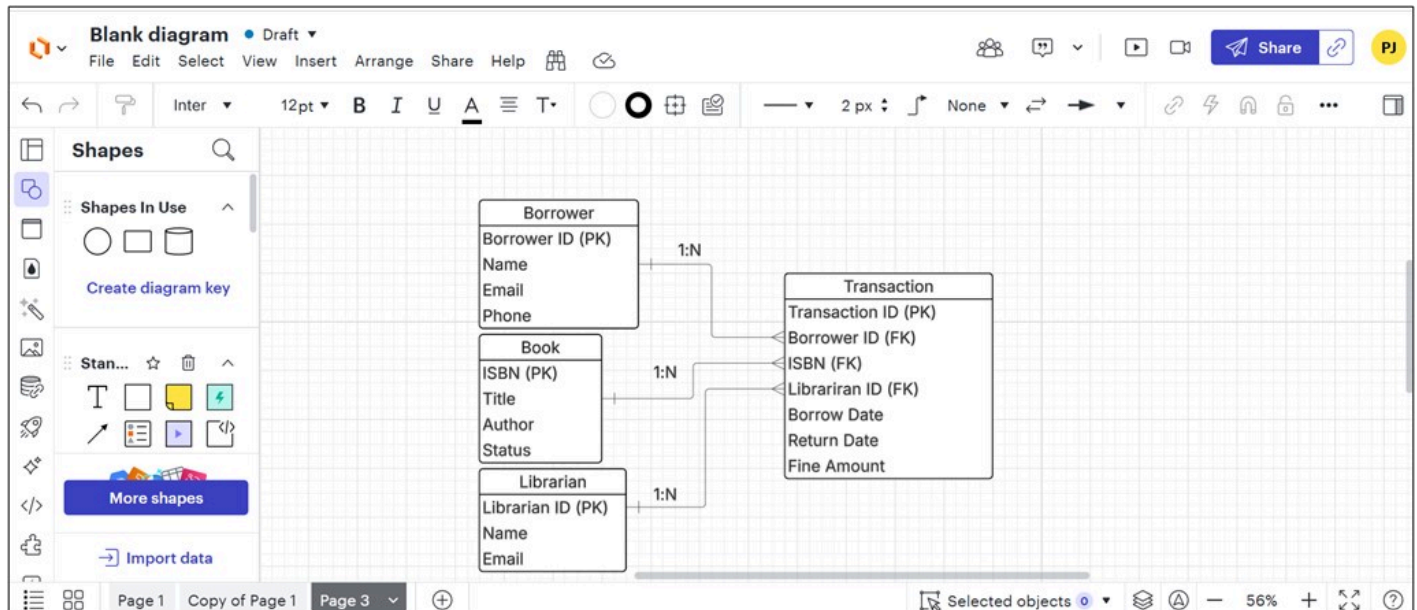
4. Normalize the database structure to at least second normal form (2NF)
5. Draw the ERD with entities, attributes, relationships, and cardinality notations

Deliverable:

A complete ERD showing:

- Four entities: Borrower, Book, Transaction, Librarian
- Attributes listed for each entity, with PKs and FKs identified
- Relationships between entities (for example, Borrower–Transaction, Book–Transaction, Librarian–Transaction)
- Cardinality notations (for example, 1:N for Borrower–Transaction)

▼ Click here to view the solution:



At this point, you have a normalized database structure that supports the system's operational needs and ensures data consistency across processes.

Summary

This lab equipped you with foundational skills to model systems and databases using DFDs and ERDs. By working through a library management system, you practiced structuring data flows and defining entity relationships. These diagrams support system analysis,

documentation, and development in real-world projects.

Author

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